

# Statistical Analysis of Spotify Song Popularity

## Project 7 Proposal and Analysis Plan

Group 3: Megan Millet, Gbenga Solomon, Carson Vrba, Priscilla Yinzime

MATH 573 – Statistical Consulting

September 2026

## **What makes a Spotify song successful?**

We will study the musical, temporal, artist, and cross-platform characteristics associated with streaming success from 2023 to 2025.

# Four Groups of Factors

We look at four groups of factors:

- **Music** – audio features like energy, danceability, tempo
- **Artist** – solo versus collaboration, artist popularity
- **Platforms** – Spotify, YouTube, TikTok, Apple Music, Deezer, Shazam
- **Time** – release year, release month, song age

All four are examined as potential drivers of **streaming success**.

# Why This Is More Than a Spotify Dashboard

## We will move from

- descriptive statistics
- visual exploration
- correlations
- statistical tests

## To

- regression modeling
- multivariate analysis
- clustering
- prediction and model comparison

**Goal:** explain patterns statistically, not simply display them.

[Spotify Songs 2023](#); [Spotify Songs 2024](#); [Spotify Songs 2025](#); [Spotify Songs 2025](#)

## **Spotify 2023 to 2025**

- streaming counts
- playlists and charts
- release information
- audio characteristics
- artist information

## **Additional platform data**

- Apple Music
- Deezer
- YouTube
- TikTok
- Shazam
- other available platform measures

**Key methodological issue:** the datasets do not have identical schemas.

# Data Integration Strategy

**Step 1.** Collect the 2023, 2024, 2025, and Wrapped 2025 datasets.

**Step 2.** Harmonize the schemas and identify comparable variables across years.

**Step 3.** Build two analytical datasets:

- a **harmonized longitudinal dataset** (comparable across years)
- a richer **platform-specific dataset** (cross-platform analysis)

Only variables that are sufficiently comparable and statistically defensible will be carried forward.

# Core Variables

## Popularity

- Spotify streams
- playlists
- charts
- cross-platform metrics

## Musical

- BPM
- danceability
- energy
- valence
- acousticness
- instrumentality

## Context

- release year
- release month
- artist count
- year of dataset
- song age

## **What characteristics are associated with Spotify streaming success?**

- ① How does streaming success differ across 2023, 2024, and 2025?
- ② Which musical characteristics are associated with higher streaming counts?
- ③ Does song age relate to streaming success?
- ④ Do collaborations perform differently from solo songs?
- ⑤ How concentrated is streaming success among artists?
- ⑥ How strongly does Spotify popularity correspond to other platforms?
- ⑦ Can song and platform characteristics explain variation in streams?
- ⑧ Can songs be grouped into distinct musical and popularity profiles?
- ⑨ Can we classify songs as high- or lower-streaming?

# From Questions to Methods

| Research question                 | Planned method                         |
|-----------------------------------|----------------------------------------|
| How are the datasets distributed? | Descriptive statistics and EDA         |
| Do years differ?                  | ANOVA and Kruskal-Wallis               |
| Which variables are associated?   | Pearson and Spearman correlation       |
| Which factors explain streams?    | Multiple regression                    |
| Are effects robust?               | Diagnostics and robust standard errors |
| Are there song profiles?          | PCA and clustering                     |
| Can popularity be classified?     | Logistic regression and ML             |
| How concentrated is success?      | Gini and Pareto analysis               |

# Step 1 – Acquire and Inspect the Data

The workflow:

**Kaggle sources → Programmatic download → Schema inspection → Data-quality report**

## Questions at this stage

- What does one row represent?
- What variables are available in each year?
- Which variables are numeric, categorical, or dates?
- Which variables can legitimately be compared across years?

## Step 2 – Clean the Data

We will address, in order:

- Missing values
- Duplicates
- Data types
- Invalid values
- Outliers
- Consistency
- Documentation

**Principle:** an extreme value is not automatically an error.

## Step 3 – Feature Engineering

We will construct variables that have direct statistical meaning.

### **Song age**

$$\text{Song Age} = \text{Dataset Year} - \text{Release Year}$$

**Streaming transformation** (streams are right-skewed):

$$Y = \log(1 + \text{Streams})$$

**Collaboration indicator:**

$$C_i = 1 \text{ if artist count} > 1, \quad C_i = 0 \text{ otherwise}$$

# What Does a Typical Hit Look Like?

We will examine:

- mean, median, variance, standard deviation
- quartiles and interquartile range
- skewness and distribution shape
- extreme observations
- differences between raw and log-stream distributions

A key issue to investigate is whether a small number of extreme hits dominate the mean.

## **Did the distribution of streaming success change over time?**

We will compare the three yearly means using both parametric and nonparametric approaches.

### **Planned analysis**

- boxplots and distribution plots
- one-way ANOVA
- Kruskal-Wallis test
- post-hoc comparisons when appropriate

# Are Musical Features Associated With Streams?

We will examine relationships between **streams** and energy, danceability, valence, BPM, acousticness, and similar features.

**Pearson correlation:**

$$r = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Spearman correlation is the correlation of ranks.

**Correlation does not imply causation.**

# Hypothesis Testing

## **Year effect**

$$H_0 : \mu_{2023} = \mu_{2024} = \mu_{2025}$$

## **Collaboration effect**

$$H_0 : \mu_{\text{solo}} = \mu_{\text{collab}}$$

## **Key and popularity**

$$H_0 : \text{key and popularity are independent}$$

We will report: test statistic,  $p$ -value, effect size, and 95% CI.

# Statistical Rigor

Because many variables will be investigated, we will address:

- multiple hypothesis testing
- effect size versus statistical significance
- assumptions of statistical tests
- nonparametric alternatives
- confidence intervals

For multiple testing, we may use Bonferroni or false-discovery-rate control.

$$\alpha^* = \frac{\alpha}{m}$$

for a Bonferroni adjustment.

# Can We Explain Streaming Success?

Our central regression model will use log streams:

$$\log(1 + S_i) = \beta_0 + \beta_1 D_i + \beta_2 E_i + \beta_3 V_i + \cdots + \epsilon_i$$

## Progressively richer models

- **Model 1:** musical features
- **Model 2:** plus song characteristics
- **Model 3:** plus year
- **Model 4:** plus platform exposure

Models will be compared using fit, parsimony, diagnostics, and predictive performance.

# Regression Diagnostics

We will not treat a regression output as the final answer.

We will investigate:

- multicollinearity
- residual patterns
- heteroskedasticity
- influential observations
- model specification
- robust standard errors

**Example diagnostic question:**

$$H_0 : \text{Var}(\epsilon_i) = \sigma^2$$

versus heteroskedasticity.

# PCA: Can We Reduce the Audio Dimensions?

Many audio variables may contain overlapping information. PCA constructs:

$$Z_1 = a_{11}X_1 + \cdots + a_{1p}X_p$$

such that the first component explains the maximum possible variance.

## Questions

- How many dimensions are needed?
- Which audio variables drive each component?
- Is there a latent musical structure?

**From many correlated variables to a smaller set of interpretable dimensions.**

# Clustering: Are There Types of Hit Songs?

After standardization, we will investigate whether songs form natural groups.

$$\min_{\{C_k\}} \sum_{k=1}^K \sum_{x_i \in C_k} \|x_i - \mu_k\|^2$$

**We will determine the number of clusters using evidence**

- silhouette score
- within-cluster variation
- cluster stability
- interpretability

*Not four clusters because four looks good, but four clusters because the data support them.*

# Is Spotify Success Universal?

For songs with cross-platform data, we will compare Spotify with YouTube, TikTok, Apple Music, Deezer, and Shazam.

**Does popularity on one platform translate to popularity on another?**

We will use correlation and multivariable models to identify which platform measures are most strongly associated with Spotify streams.

# Is Streaming Success Concentrated?

We will examine whether a small number of artists or songs account for a disproportionate share of streams.

## Tools

- artist-level aggregation
- Pareto analysis
- concentration curves
- Gini coefficient

The Gini coefficient is:

$$G = \frac{\sum_i \sum_j |x_i - x_j|}{2n \sum_i x_i}.$$

# Can We Identify a High-Streaming Song?

Define:

$$Y_i = 1 \text{ if } S_i \geq Q_{0.75}, \quad Y_i = 0 \text{ if } S_i < Q_{0.75}$$

where  $Q_{0.75}$  is the 75th percentile of streams.

**Logistic regression:**

$$\log \left( \frac{p_i}{1 - p_i} \right) = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_k X_{ik}.$$

We will interpret coefficients using odds ratios:

$$OR_j = e^{\beta_j}.$$

# Prediction Is Not the Same as Explanation

We will separate:

- **Explanation** – statistical inference
- **Prediction** – out-of-sample performance

Predictive models will be evaluated on held-out data using:

accuracy, precision, recall,  $F_1$ , ROC-AUC.

# 11-Week Project Plan

| Weeks | Planned work                                                 |
|-------|--------------------------------------------------------------|
| 1–2   | Data acquisition, schema inspection, cleaning, harmonization |
| 3     | Feature engineering and master analytical datasets           |
| 4     | Descriptive statistics and exploratory data analysis         |
| 5     | Temporal, artist, and release analyses                       |
| 6     | Correlation analysis and hypothesis testing                  |
| 7–8   | Multiple regression, diagnostics, and robust inference       |
| 9     | Cross-platform analysis and artist concentration             |
| 10    | PCA, clustering, and cluster validation                      |
| 11    | Logistic regression, prediction, and final integration       |

## Deliverables by Phase

| Phase                  | Weeks | Deliverable                                  |
|------------------------|-------|----------------------------------------------|
| Phase I: Preparation   | 1–3   | Clean, harmonized, engineered datasets       |
| Phase II: Exploration  | 4–5   | EDA report and temporal/artist trends        |
| Phase III: Inference   | 6–8   | Correlation, hypothesis tests, regression    |
| Phase IV: Multivariate | 9–10  | Cross-platform, concentration, PCA, clusters |
| Phase V: Prediction    | 11    | Classifier, final report, presentation       |

**Each phase produces an auditable result before the next phase begins.**

# How We Will Work as a Group

## Shared responsibilities

- agree on research questions
- maintain one data dictionary
- use reproducible code
- document every preprocessing decision
- review each other's analyses

## Reproducibility

- programmatic data acquisition
- fixed random seeds where needed
- version-controlled analysis
- saved cleaned datasets
- documented statistical decisions

**No analysis without documenting why the method was chosen.**

# What We Hope to Learn

- ➊ **Musical characteristics** – which audio features are associated with streaming success?
- ➋ **Time** – has the profile of popular songs changed from 2023 to 2025?
- ➌ **Platforms** – is popularity transferable across digital platforms?
- ➍ **Artists** – how concentrated is streaming success?
- ➎ **Statistical structure** – are there identifiable groups of songs?
- ➏ **Prediction** – how accurately can observable characteristics identify high-streaming songs?

# Limitations We Will Address

- Kaggle datasets may represent selected popular songs rather than all Spotify songs.
- Dataset schemas differ across years.
- Cross-platform variables are not available for every dataset.
- Observational data support association, not automatic causal claims.
- Streaming counts may reflect exposure, promotion, and artist popularity.
- Differences in data collection may affect comparisons across years.

**These limitations will be treated as part of the statistical analysis.**

# From Raw Data to Statistical Conclusions

**Raw data → Clean → Harmonize → Engineer → Explore → Infer → Model → Validate  
→ Interpret → Communicate**

**The objective is not to find the most impressive graph.**

**The objective is to answer meaningful questions with defensible statistics.**

# The Project in One Sentence

We will investigate how musical, temporal, artist, and cross-platform characteristics are associated with Spotify streaming success using a reproducible statistical workflow from data cleaning through inference, modeling, clustering, and prediction.

# Thank You

Questions and suggestions